

Chapter One

Healthy Life, Better Living



We cannot live without food. Several types of food are regularly necessary for the growth and development of the body, repair and replacement of worn-out tissues, and for producing energy etc. Our health depends largely on the quality of the food we take. Food can affect in our appearance, work, behaviour and standard of life. During the process of respiration the chemical energy of food is converted into thermal energy and other forms of energy. Every living being takes food from the environment according to its need. Each type of food is a complex chemical compound. This complex food is broken down into simpler substance by different types of enzymes in different parts of our digestive system. This process is called digestion. The digested food is then absorbed into the protoplasm of body cells. This is called assimilation. After digestion the indigestible part of the food excreted from of the body through a special process.



At the end of this chapter we will be able to-

- explain the components of food and an ideal food pyramid;
- describe the needs of the preservation of food;
- analyse the impact of natural and fast food on the preservation of health;
- explain the sources of vitamins and their deficiency symptoms;
- explain the sources of minerals and their deficiency symptoms;
- explain the necessity of Body Mass Index (BMI);
- explain the usefulness of water and fibrous food;
- describe the use of chemicals in food and their effects;
- explain the harmful reactions of tobacco and drugs in human body;
- explain what AIDS is;
- explain the techniques of maintaining physical fitness.

1.1 Food and Nutrition

According to the nutrition science, all that we eat is not food. The edible items that help produce energy, support growth, repair and immunity- in short, nourish the body- are called food. Nutrition is- getting necessary food items from the environment and digesting and absorbing them, then fulfilling the energy need, developing the immunity of diseases, and supporting the growth of the body. On the other hand, the organic and inorganic elements in food which provide the energy required for living body are collectively called the nutrients e.g. glucose, minerals and vitamins etc. The three main functions of food are-

1. growth and development, repair and protection of the body.
2. production of heat and energy.
3. protecting the body from diseases and keeping the body healthy, energetic and active.

Elements of Food

There are six components of food (Fig: 1.01) e.g.- carbohydrates, proteins, fats, vitamins, minerals and water. In addition, another important component of food is fibre. Fats and carbohydrates are called the energy-producing foods and proteins are called the body building foods. Although these components have other important functions. Vitamins, minerals, water and fiber help preserve the living processes of the body in many different ways.

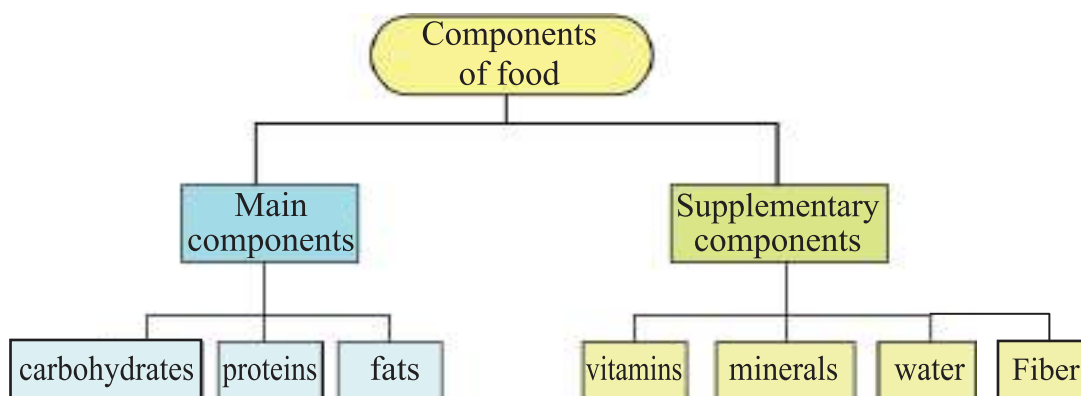


Fig. 1.01 : Components of food

1.1.1 Carbohydrates

Carbohydrates are the main food for human beings. They are composed of carbon, hydrogen and oxygen. They produce energy and heat. Sugar has no colour and smell but is sweet in taste. There are several types and sources of carbohydrates.

Plant Sources

Starch: Rice, wheat, maize and many other cereals are the main sources of starch. Besides these, potato, sweet potato and arum are vital sources of starch.

Glucose: It is less sweet than sugar. This carbohydrate is available in grapes, apples, carrots, dates etc.

Fructose: In mango, papaya, banana and orange, or other sweet fruits and in flower-extracted honey, fructose is available. This is called fruit sugar.

Sucrose: Sugarcane, sugar, molasses and sugar candy are the sources of sucrose.

Cellulose: Wood apple, mango, banana, watermelon, groundnut, dry fruit and all kinds of vegetables contain cellulose.

Animal Source

Lactose or Milk Sugar: The milk of cows, goats and other animals are the sources of lactose.

Glycogen: Glycogen is available in the meat and liver of birds like chicken, pigeon etc.

Nutritional Value

Carbohydrate has an important role in nutrition. Carbohydrate produces energy and heat in body. During respiration carbohydrate is oxidized and produces energy for metabolism. Glycogen provides energy in deficiency of food or in hard labour. Cellulose is a non digestive type carbohydrate. This is a fibrous food. It helps in regular bowel movements and protects the body from constipation. Besides, proteins and fats are synthesized from carbohydrates.

To get rid of the sugar deficiency diseases, we have to take an optimum amount of sugar every day. On the other hand if the amount of sugar is more than the need, extra sugar is stored as fat in the body. It may cause obesity and sometimes even diabetes.

Through the breathing or respiration process, we inhale oxygen which gets mixed with blood in our lungs. Red blood cells of blood carry this oxygen to the cells of our body and there it reacts with glucose to produce different forms of energy including heat. This energy is the source of all our energies.

The energy in food is measured as food calorie or kilocalorie. The amount of heat energy produced by the oxidation of one gram of food is called the food calorie. The amount of energy which is needed to increase 1° Celsius temperature of 1 gram water is called one calorie. One thousand calorie is equal to one kilocalorie or one food calorie. Food calorie is expressed in kilocalorie. The calorie of carbohydrates and protein is almost equal 4 kcal/g, while fat has the highest amount of calorie, that is, 9 kcal/ g. An adult man needs to eat food approximately equivalent of 2500 kcal per day and for a woman this figure is approximately equivalent of 2000 kcal a day.

1.1.2 Proteins

A protein is composed of carbon, hydrogen, oxygen, nitrogen and sometimes sulphur. A protein is converted into an amino acid after digestion. So far 20 types of amino acids have been found in the human body. Amino acids are the units of protein formation.

According to their sources, there are two types of proteins- animal proteins and plant proteins. The proteins which are obtained from animals are called animal proteins e.g. fish, meat, egg, milk, cheese etc. The proteins which are obtained from plants are called plant proteins. e.g. pulse, bean, seeds, pea, nuts, etc.

Nine out of a total of 20 amino acids are essential. These are lysine, tryptophan, methionine, valine, leucine, isoleucine, histidine, phenyle, alanine and threonine. Body can synthesize the other amino acids except these types. These types of proteins are more available in animal protein. That is why the nutrient value of animal protein is much more than plant proteins. Plant foods like pulse, soybean, pea and maize are rich in higher nutrient value. The other

plant foods have no essential amino acid. So these plant foods have less nutrient value.

Proteins are essential for the building of animal body. Most of the body are formed of proteins. Bone, muscle, hair, bird feather, nail, animal horns etc. are all made of protein about 50% of the dry weight of an animal cell is protein.

1.1.3 Fats and Lipid

Fats are composed of fatty acids and glycerol. There are 20 types of fatty acids in food. The quality of fats depends on fatty acids. Solid lipids are called fats. Lipids are saturated fatty acids. They are solid at normal temperature and pressure e.g. the fats of fish and meat. The lipids which are liquids at normal temperature and pressure are called oils. Oils are unsaturated fatty acids. They are liquid at normal temperature and pressure e.g. soybean oil, mustard oil etc. Lipids are of two types according to their sources.

1. **Animal lipids:** Meat with fat, butter, ghee, cheese, yolk of eggs are the sources of animal fat.
2. **Plant lipids:** Different types of vegetable oils are the sources of plant lipids. Mustard, soybean, sesame, linseed, maize, coconut, sunflower and palm oil are the sources of plant lipids. Cashew, pistachio and peanut are also the sources of lipids.

Role of Lipid

1. Lipid produces the highest amount of heat and energy among all food components.
2. Lipid is essential for the growth and nutrition of the body.
3. It prevents the misuse of heat and works as a source of food storage for the future.
4. It keeps the skin smooth, soft and healthy and thus protects the skin from skin disease.
5. Lipid helps to absorb the soluble vitamins e.g.- A, D, E and K.

Diseases Resulting from Lipid Deficiency and Remedies

The deficiency of lipids causes skin to become dry and rough which damages its natural beauty. The long time deficiency of lipid causes break down of the stored protein and body loses its weight. On the other hand too much of lipid disturbs the blood circulation of the body. Fatty body is more prone to suffer from different diseases.

1.1.4 Vitamins

Even if the food contains sufficient amounts of carbohydrates and proteins, a special type of food is necessary for the normal growth and nutrition of the body. This food component is called vitamin.

Vitamin is essential for normal growth and a healthy body. Vitamin is an organic compound. If we eat a healthy diet with adequate vegetables everyday, we can get the required vitamins from our daily food. Some vitamins are soluble in fats and some are soluble in water.

Fat soluble vitamins : Vitamin A, D, E and K.

Water soluble vitamins : Vitamin B complex and Vitamin C.

Fat Soluble Vitamins

Vitamin A

Animal sources of vitamin A are eggs, cow's milk, butter, cheese, yogurt, liver and different types of oil rich fish, specially the cod fish. Plant sources of vitamin A are carotene rich vegetables e.g. red amaranthus, arum leaves (kachu shak), stem amaranth (data shak), basil, jute leaf, water spinach (kalmi shak), mint, pea, carrot, lady finger, cabbage, peanut and different types of fruits e.g. mango, ripe papaya, jackfruit are the great

sources of vitamin A. Large amounts of vitamin A is present in carrots. The functions of vitamin A are:

1. It ensures smooth activities for the growth and development of the body.
2. It maintains normal activities of different types of epithelial tissue e.g. skin and cornea of the eyes.
3. It keeps teeth, gum and bones healthy.
4. It keeps the eye sight normal and protects the eyes from night blindness.
5. It protects the body from contagious diseases.

Deficiency Symptoms of Vitamin A and Remedies

Vitamin A deficiency causes xerophthalmia. Although this condition initially presents as night blindness, prolonged deficiency can lead to ulceration or erosion of the cornea of the eye. The affected person can become completely blind if this disease occurs. In the absence of vitamin A, symptoms such as sores, colds, coughs, sore throats, urinary problems etc. are also seen. A lack of vitamin A can cause small bumps at the base of hair follicles on skin. Taking too much vitamin A can also be harmful.

Vitamin D

Animals are the only source of vitamin D. Vitamin D is synthesized in human skin with the help of ultraviolet rays, from the cholesterol present in the human skin that is transformed into vitamin D step by step through the liver and kidneys. Yolk of eggs, milk and butter are the main sources of vitamin D. Cabbage, liver and oil rich fish are also the sources of vitamin D. Vitamin D helps the body absorb calcium which is useful to build bones. Because of the deficiency of vitamin D, children may be attacked with rickets disease. Taking more vitamins than the need is harmful to the body. So, a huge amount of calcium and phosphorus are absorbed. As a result, a large amount of calcium is stored in the kidney, heart and veins.

Vitamin E

All types of vegetable oil, especially palm oil, are the best sources of vitamin E. Almost all food contain some amount of vitamin E. Vitamin E is also found in corn oil, cottonseed oil, sunflower oil and lettuce leaves. In human bodies vitamin E acts as the anti oxidant. This anti oxidant inhibits the accumulation of fat in arteries and keeps the skin healthy. Eating a balanced diet everyday can fulfill the demand of vitamin E.

Water Soluble Vitamins

Vitamin B complex

There are 12 types of water-soluble vitamin B. This group of vitamins is called the vitamin B complex. Vitamin B complex is very important for normal health. The presence of vitamin B complex is essential for the growth of the body especially for the work of the brain and the nervous system, metabolic activity of the cell and for reproduction.

The sources, deficiency symptoms and the amount of different types of vitamins in the B complex group are shown below:

Vitamin	Source	Symptoms
Thiamin (B ₁)	Plant source: husked rice, flour, pulse, oil seed, nut, fresh fruits and vegetables. Animal source: Liver, egg, milk, fish etc.	Severe deficiency of vitamin B1 causes the symptom of beriberi disease. Weak nerve, depression, tiredness, loss of appetite, weight loss are also caused by its deficiency.

Riboflavin (B ₂)	Liver, egg, milk, green vegetables, young shoot of plant and germinated seed.	Its deficiency causes cracks at the corners of the lips, sores in the mouth and on the tongue, and rough skin.
Niacin or Nicotinic acid (B ₃)	Meat, liver, flour (ata), pulse, nut, oil seed, chick-pea and vegetables.	Its deficiency causes pellagra disease. The main features of pellagra are dermatitis (inflammation of skin), diarrhea and dementia (memory problem), collectively known as 3D.
Pyridoxine (B ₆)	Rice, ata, fish, meat, vegetables, chick-pea, fungi, kidney and yolk of eggs.	Loss of appetite, vomiting tendency and anaemia occur due to its deficiency.
Cobalamin or Cyanocobalmin (B ₁₂)	Liver, milk, fish, meat, egg, cheese kidney etc.	Anaemia and nervous system disorder occur due to its deficiency.

Vitamin C (Ascorbic acid)

Fresh vegetables and fruits contain vitamin C. The fruits such as gooseberry (amloki), lemon, orange, tomato, pineapple and guava are the sources of vitamin C. The vegetables such as radish, lettuce, coriander leaf, mint, green chili, cauliflower and bitter gourd contain large amounts of vitamin C. Dry fruit, seed and canned food usually do not contain this vitamin.

The roles of vitamin C are-

1. The cells of skin, bone, teeth become compact and strong.
2. The damaged cells get repaired.
3. Teeth and gum become strong.
4. Vitamin C has an important role in the metabolism of fat, protein and amino acids.
5. It keeps the skin bright and smooth.
6. It protects the body from diseases.

Due to its severe deficiency scurvy disease (bleeding from the gum of teeth) occurs. For its deficiency (i) Bones cannot become strong. (ii) Sores appear on the skin and wounds take time to heal. (iii) The gum of the teeth swells and teeth become weak. (iv) The immunity of the body decreases and frequently catches cold.



Individual Work

Task : Make a chart of the diseases caused by the deficiency of vitamins that we have discussed.

1.1.5 Minerals and Water

For the normal growth and nutrition of the body, minerals are as essential as vitamins. Minerals mainly help in the formation of cells. Animals get minerals from the plant source. We fulfill the demand of minerals by eating vegetables, fruits, milk, egg, fish and drinking water. The source, nutritional value and the deficiency symptoms of some important minerals are described below:

Iron (Fe)

Iron is the main element of blood. 100 mL blood contains 50 mg iron. It is stored in liver, spleen, bone and red blood cells. The plant sources of iron are

cauliflower, spine amaranths, neem leaf, fig, green banana, maize, wheat, nut and millet. The animal sources are fish, meat, egg and liver. The main function of iron is hemoglobin formation. Hemoglobin deficiency causes anaemia. The symptoms of anaemia are pale eyes, swelling of hands and legs, weakness, headache and palpitation.

Calcium (Ca)

The main element of bone and teeth is calcium. About 2% of human body weight is calcium. The amount of calcium is the highest among the minerals. Of them, 90% is accumulated in the bone and teeth accompanied with phosphorus and magnesium. The amount of calcium is significant in blood and lymph. The plant sources of calcium are: pulse, sesame, soybean, cauliflowers, carrots, spinach, kochu leaf, red amaranthus and kalmi leaf. The animal sources of calcium are milk, eggs, small fish and dry fish.

Calcium is essential for strong bone and teeth. Besides this, calcium helps in blood circulation, proper contraction of the muscle of the heart and the movement of nerve and muscle. Rickets and osteomalacia of aged women occur due to deficiency of calcium. Calcium deficiency delays in the formation of teeth of babies and disturbs in blood circulation.

Phosphorus (P)

The importance of phosphorus regarding its amount in the body is next to calcium. Like calcium, phosphorus is also an element of the bone. Phosphorus is accumulated in the bone, liver and plasma. Phosphorus plays a vital role in the formation of nucleic acid and nucleoprotein and in producing energy through carbohydrate metabolism. Plant source- cereal, beans, peanuts and nuts are the sources of phosphorus. Animal source- Eggs, milk, fish, meat are the sources of phosphorus.

Like Calcium, the main function of phosphorus is the formation of bone and teeth. Osteoporosis, rickets and dental caries result from phosphorus deficiency.

If there is sufficient amount of protein and calcium in food, phosphorus deficiency does not occur.

Water

Water is one of the main components of food. Water is essential for the human body. The structure and function of the body cannot be maintained without water. Water constitutes 60-75% of our total body weight. Water is essential for the formation of our blood, muscle, nerve, teeth, bones etc.

It is not possible to form the body cell and maintain physiological function of the body without water. It works as a solvent in the human body. Water helps in digestion and absorption of food. Metabolism produces urea, ammonia etc in our body. Water helps to dispose these toxic substance from the body through urine and sweat. Besides this, water keeps our body cool by sweating and evaporation.

The sources of water in body-

1. Drinking water, other drinks e.g. tea, milk, coffee and juice.
2. By taking food e.g. vegetables and fruit.

If the intake of water and release of water are equal, the balance of water is maintained in the body. An adult person should drink 2-3 liters of water per day.

Hot weather and hard labour are the causes of deficiency of water in the body. So, the amount of drinking water should be increased in this situation. If diabetes is not in control, it causes repeated urinating. So, the deficiency of water occurs in the body.

Severe thirst, inhibition of blood circulation and shrunk skin result from the deficiency of water. The nerve and muscle also become weak due to lack of water. The balance of acid and alkali is lost and it causes acidosis. 10% of water loss of the body causes fainting and even death of a person. Excessive vomiting and diarrhoea can also lead to water loss from the body. The patient should take salt water or saline water for the rapid cure of water deficiency. The water

and salt which come out of the body, are recovered by saline water. If oral saline is not available in the house, it can be made by mixing a pinch of salt, a handful of molasses or sugar and a glass of boiled safe water at room temperature.

1.1.6 Fibre

Besides the components of food, discussed so far, another very important component of food is roughages or fiber-rich foods. Fibre or roughage is mainly obtained from plants. Entire seeds, pulse, potato, fruits with peel and vegetables are the sources of roughage. Besides this, dry fruit, cumin, coriander and pea contain a large amount of roughage. Roughage is mainly cellulose contained in the plant cell wall. Roughage does not supply any nutrient to our body. It can protect the body from constipation, heart disease and diabetes. How roughage protects these diseases are yet to be known clearly. Roughage can go right through the intestine. It is not digested.

The importance of roughage food is as follows:

1. It helps to digest food. It absorbs water and increases the amount of stool.
2. It helps to dispose the undigested food from the body.
3. It helps to decrease the extra fat of the body.
4. It also decreases the tendency of frequent hunger.
5. It is thought that roughage food can decrease gallbladder diseases, cancer of the intestine and colon, piles, appendicitis, heart diseases and obesity.

So we should take 20-30 grams of fiber food everyday. This fiber is obtained from vegetables and fruits.

1.2 Body Mass Index (BMI)

A baby continues to grow after birth. Then it becomes a child, reaches teenage, youth and adulthood. The growth of human body continues up to the age of 20-24 years. The height does not increase after that time. Then the role of food

is to repair and keep the body healthy. In adult age, for good health, it needs to maintain a balance between height and weight of the body. The index which shows the balance in height and weight is called the BMI or Body Mass Index. If the body weight is balanced with the height, then it can be considered a nutritionally healthy body.

The law of BMI

$$\text{body weight (kg) / height (meter)}^2$$

That is, the body weight of a person is divided by the square meter of his height and that is the BMI of that person.



For example:

The body weight of a person is 80 kg and height is 1.8 meter.

$$\text{BMI} = \frac{80}{1.8 \times 1.8} = 24.7 \text{ (approximate)}$$

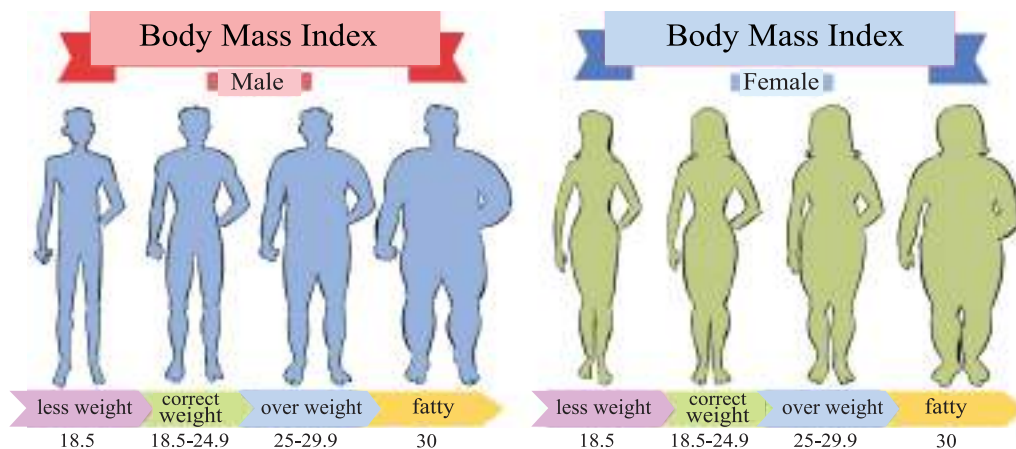


Fig. 1.02 : Body Mass Index

BMI is the indicator of fat in our body. Fig. 1.02 shows that 18.5-24.9 is healthy and normal BMI. If a person has less or more BMI than this, he/she is considered under-weight and obese respectively. Note that BMI is not the only or most accurate indicator of health, but it is widely used.



Group Work

Calculate the BMI of all the students of your class and show it in a graph chart (Fig : 1.03). What is the average BMI of your class?

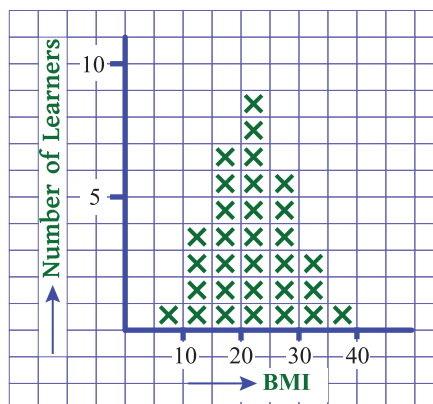


Fig. 1.03 : BMI and group of learners

1.3 What will be the daily diet

We have learnt about the calorie and kilocalorie of food while discussing the nutritional value of food in this chapter.

An adult, who does physical work, should take food of 2000-2500 kilocalories per day. It is necessary to eat vitamins, minerals and fibrous food along with vegetables and fruits. (The fibre element of food is not shown separately in this figure.)

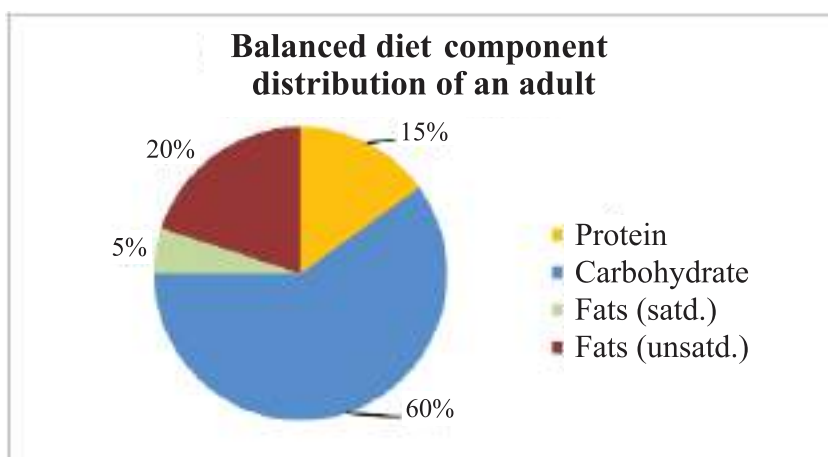


Fig. 1.04 : Balanced diet component distribution of an adult



Individual Work

Task : Hilsa fish, chicken egg, meat with fat, bean seed, yogurt, rice, potato, sugar, oil, sweet pumpkin, cauliflower, tomato, small fish, chick-pea, ice cream, bread, honey, ghee, basil, jackfruit, mango.

Make a chart of food elements with the 21 food items mentioned above. Same food can be part of more than one list.

Carbohydrate	Protein	Fat	Vitamins and minerals	
			Vegetables	Fruit

Now, sort out the foods as low cost and high cost. According to the table below make a list of low cost and high cost foods.

List of the foods (low cost food)

Name of the food components	Low cost food	Costly food
1. Carbohydrate		
2. Protein		
3. Fats		
4. Vitamin enriched vegetables/fruits		
5. Mineral enriched vegetables/fruits		

From the above table you can surely assume that to eat healthy food it is necessary to eat costly food² is not true. By proper planning we can eat healthy food with minimum cost.

1.3.1 Balanced Diet

We have already come to know what food is and what the elements of food are. Over-eating is harmful to health and at the same time eating less food is also harmful to health. Becoming obese by eating more is a big problem in the first world. So, we have to take balanced food for good health.

Balanced diet does not mean a specific food. Balanced diet is a diet which contains all the seven elements of food in a specific amount and eating this food we get the necessary calories and fibres for our normal activities. For example, an active, healthy adult male needs about 2500 and an average female needs about 2000 kilocalorie energy daily. We get this calorie from food. So, in our daily food list we need to select such food items in which all the seven elements of food are present in a proper amount.

For preparing a list of balanced food we have to consider the age, sex and physical work done by the person. For the preparation of the list of food for a baby and an aged person, easily digestible and fat free food should be considered. Growing babies need protein, calcium and phosphorus-enriched food for the growth of bones and teeth. Extra proteins, calcium, and iodine are very important for producing blood and for the development of the embryo in a pregnant person. No definite balanced food is available in nature. We have to prepare the balanced food.

The preparation of balanced food

Carbohydrate	Protein	Fat	Vitamin	Mineral
Rice	Fish	Butter	Milk, Eggs	Milk
Wheat	Meat	Oil	Fruit	Eggs
Sugar/ Molasses	Eggs	Ghee	Fish/Meat	Vegetables

The Pyramid of Balanced Diet

It is important to include carbohydrates, vegetables, fruits, proteins and fats in a balanced diet menu. It is seen that the amount of carbohydrate is highest in the menu of a teenager or an adult. In food pyramids of a balanced diet, carbohydrates are placed at the bottom of the pyramid because the amount of carbohydrates is higher than the other elements of food. Then the vegetables, protein and fats are placed one after another in a pyramid (Fig : 1.05). So the fats are placed on the top of this pyramid.

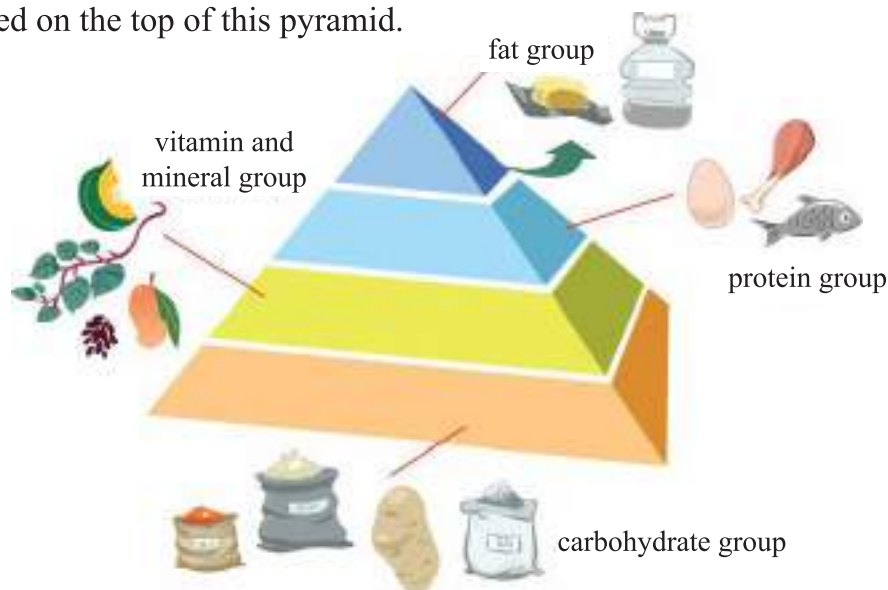


Fig. 1.05 : Pyramid of balanced diet

1.3.2 Selecting Food for High Standard Living

The food habits of all men are not the same. The availability of food is not same in all countries due to geographical and natural causes. The need of foods is different in winter and in summer to adjust with life. Selection of food depends on growth and development and physiological activities of the body. So, food should be selected by considering the amount and its calorie value for high standard of living.

In the food pyramid, we have shown how much of which type of food component should be eaten. Now, we will say what foods can be made using these elements of food. For example, oil or butter shown in the pyramid are not eaten by themselves, they are used to make other foods.

Nutritionists have divided the sources of nutrition into four classes. These are :

1. Meat or fish, eggs, pulses (peanuts, chick pea and nuts).
2. Fat or lipids
3. All kinds of edible vegetables and fruits.
4. Cereals and their products e.g. in bread and rice.

To get balanced food we should take all these four types of food every day. Nutritionists think that the selection of food should have a variety from these four classes.

The amount of proteins, carbohydrates, fats, vitamins and minerals should be considered for the preparation of the menu from breakfast to dinner. In our country, breakfast completed is with light food. Taking tea is a habit among the aged people of towns and villages. Many of the city dwellers complete their breakfast just with a cup of tea. This habit is very harmful to the health. It is necessary to eat at least some snacks with tea. All the nutrients will be easy to get if the break-fast is completed with bread, butter or an egg and a banana. In summer, body remains healthy if the breakfast is done with molasses of sugarcane and flakes rice (chira and goor). Lunch is considered the main meal in our country. While selecting the food items for lunch one must plan the meal based on a balanced diet chart.

In the tropical regions (tropical countries) fish is an excellent source of protein. However, in winter, including meat along with fish adds variety to the meal. Yogurt and fruit after every meal are good for the health.

In our country, many students and service holders have no specific time for lunch. So, they take a light food in the evening. The evening snack should be prepared according to the economic condition of the family as well as the nutrient value of the food.

The food for dinner should be easy to digest. So less amount of protein should be taken in this meal. A high standard of living can be maintained by this type of food selection.

Fast Food or Junk Food

Fast food or junk food is such a food which is tasty rather than nutritious. These foods are very tasty to eat but not good for the health. To enhance taste, they are often mixed with excessive chemical substances that are unhealthy. There are lots of animal fats and sugar in these foods. Burger, fried chicken, pizza, chips, chrispy foods, cake and biscuit have lots of animal fat. There is a high amount of sugar in soft drinks and in carbonated drinks like cola or lemon. When we take these fat foods, it turns into fat cells in our body. Taking much sugar can damage our teeth and skin. Fast food can never be a balanced food. They lack vitamins and miner-als which are essential for our body. Fast food is the cause of obesity for growing boys and girls. In fact, fresh natural food is better than canned or packet foods.



Individual Work

Task : Make a list of the food components that are present and absent in fast food

1.4. Preservation of Food

All types of food get spoilt or become unsuitable for eating due to natural causes. The causes of food spoilage are- getting attacked by germs and fungi, their quick multiplication, increase of enzymes in the food, humidity of the environment and increase of acidity due to rise of temperature. None of these causes alone spoils food. Several reasons together spoil food. For example, the increase of humidity in the environment helps multiply germs and spoil food. Similarly, the rise of temperature increases the amount of enzymes in food and spoils it.

Germs or bacteria produce toxic elements in the food. These toxic elements are called toxin. This toxin is of different types. Being affected by some of these toxins is called food poisoning. Some toxins attack the nervous system and may cause death.

Yeast, a kind of fungus, quickly destroys the fruit juice, tomato sauce, jelly, sweet pickle and sarbot. So the food gets a bad sour smell. If bread is kept in an open place for few days, an ash colour layer grows on it. These are the mold fungus like mucor and aspergillus. Oranges, pickles, tomatoes and cheese rot due to this type of fungus.

By storing food, we can have the taste of some seasonal fruits, crops, vegetables, fish and other foods in other seasons. If the production of any fruit or crops is very much at a particular time or place, we can consume, shift and export them to other places at different times through preservation. So, we should preserve various kinds of food by different processes maintaining the proper food value. Thus, we can meet up our food deficiency.

1.4.1 Various Methods of Food Preservation

Actually foods are spoilt by the multiplications of germs and the enzyme secreted by the germs. Humidity and heat help germs grow and increase the activities of the enzyme. As a result, this condition helps spoil the foods. If the factors which help ruin the foods can be controlled, then foods can be preserved for a long time.

Commercially foods are preserved and marketed adopting special methods. At home foods are preserved by using preservatives and machineries. Some of such methods are described below.

1. Drying: Drying is an ancient method for preserving food. Fungus and bacteria can not attack the dry food. So, the activities of enzyme are inhibited. Foods are preserved for a long time by this method.

2. Refrigeration: Mainly vegetables, fruit, cooked food and sweetmeats can be preserved by the refrigeration method for several days. In this process the multiplication of germs and the activities of germs cannot be prevented for a long time.

3. Freezing: In this method foods are kept at a temperature of 0° Fahrenheit or minus 18° Celsius or below so that the foods remain in good condition for a long time. Not only the fresh vegetables, fruits, juice, fish and meat but also ready made food and ice cream are preserved by applying this method.

4. Preservatives: The chemicals which are used to preserve food are called preservatives. Chemical preservatives are used to prevent food spoiling and to hinder the growth of fungus and bacteria. These preservatives have no nutritional value. The preservatives should be used in a specific amount. There are various types of preservatives and their uses are also different. Fungi and bacteria cannot grow in food due to use of preservatives.

Some important preserva tives are mentioned below:

- i) Vinegar is a common preservative. It is used in pickles and sauces. 4%-10% aqueous solution of acetic acid is called vinegar.
 - ii) Sulfate salt e.g. sodium bisulfate or potassium meta-bisulfate are used to prevent the multiplication of fungus, bacteria and other micro organisms.
 - iii) Sodium Benzoate: This is a salt of benzoic acid. It specially prevents the fungal multiplication and is used for preservation of fruit juice and fruit pulp.
- Besides these, salts of Propionic Acid and Sorbic Acids are used for the preservation of yogurt, sweets, cheese, butter and bakery food items.

The amount of preservatives mentioned above is different for different foods. Preservatives should be used in the proper quantity. Otherwise it becomes harmful to health.

5. Preservation in Sugar or Salt Solution: Salt and sugar solutions have been used for preservation for a long time. The salt solution is called the brine. Sugar and salt eases the exosmosis of microbes so that foods are not rotten- jam, jelly and marmalade are preserved in sugar. The pieces of guava, apple, and pineapple are preserved for a time in the air tight concentrated solution of sugar.

Before using the preserved food, we should be careful. If the food colour changes, food swells up, if any white or black layer is noticed and the surface of the food becomes slippery, then it should be understood that the foods have started rotting. One should avoid taking this kind of food because it may cause food poisoning.

1.4.2 Use of Chemicals for Preservation of Foods, and its Physiological Effects

A toxic chemical called formalin is often used to preserve milk, fruit, fish and even meat from rotting, although there is considerable doubt as to whether it has any role in food preservation at all. Use of formalin is prohibited for preservation of food. Some dishonest businessmen use formalin for food preservation. Long-term use of formalin causes indigestion, diarrhoea, asthma, damage of the liver and kidney or cancer. Due to long-term use of formalin, child-bearing mothers may give birth to malformed babies.

The chemicals named Ripen and Ethylene are used for the quick ripening of mangoes, tomatoes, bananas or papayas. If ethylene is used on fruit, these should be sold in the market after 7-8 days of its use. But this is not done and fruit is supplied to the market within 2-3 days. So, the effect of chemicals remains and causes diseases in the human body. Besides this, calcium carbide is used for ripening fruits. This calcium carbide forms acetylene gas in the presence of air vapour. Then this acetylene gas is transformed into acetylene ethanol which is very harmful to health.

A hormone called culter is used for the delayed ripening to mangoes. Culter is used as spray on the mango trees. Consequently, mango cannot ripen quickly and can stay on the tree for a long time. Culter is also very harmful to health.

For the prevention of the use of these toxic chemicals, the consumers' right law should be applied strictly. So the printing media and electronic media can circulate this information to raise consciousness in the society so that the common people do not buy this fruit. The people, who are involved in this dishonest activity, should be punished by the government. The mobile court and the consciousness of the people can be very effective in this regard.

1.5 Tobacco and Other Addictive Substances

The leaf and branch of tobacco plants are dried to make tobacco. Dried tobacco leaves are cut into small pieces and wrapped with paper to make cigarettes, biri and churut. Burning these and inhaling the smoke and vapour is called smoking. The nicotine of tobacco temporarily stimulates the nerve as well as endangers the body. Besides nicotine, other toxic chemicals also enter into the body by smoking. Cigarette smoke contains some toxic gases, chemicals and narcotics. These substances decrease the oxygen carrying capacity of hemoglobin in the blood. Besides, there are some sticky substances and hydrocarbons which cause various lung diseases (Fig: 1.06) and even cancer.

1.5.1 Bad Effects of Smoking

Smoking is the most familiar drug to us. The harmful situations and diseases resulting from smoking are shown below:

- i) Smokers are attacked with diseases more frequently than others.
- ii) Smokers suffer from one or the other diseases such as lung cancer, lip, mouth, laryngs, throat and urinary bladder cancer, bronchitis, ulcer in stomach, heart and blood-related diseases. Most of the patients of lung cancer die within about five years.
- iii) Studies show that the life expectancy of the smokers shortens.
- iv) The people, who do not smoke but stay near the smokers and take smoke indirectly, are also affected.



Fig. 1.06 : Lungs of a non smoker (left) and smoker (right)

1.5.2 Attempts to Control Smoking and Use of Tobacco Products

- i) Smoking is prohibited in buses, trains, open fields, restaurants, offices, hospitals and rail stations. Smoking in public places is a punishable offence. There is a law in our country against this. But it is a matter of regret that there is no enforcement of this law and people smoke anywhere they like and pollute the surroundings. This law should be strictly enforced. The Government should take bold steps for anti-smoking campaign.
- ii) It is mandatory to print horrific images of smoking-related diseases along with the sentence 'Smoking is like taking poison' or 'Smoking is harmful to health' on the packets of the tobacco products.
- iii) Advertising of tobacco or tobacco products has been banned.
- iv) It is imperative to strictly prohibit selling and advertising of cigarettes and tobacco products near schools, colleges or other educational institutions.

1.6 Drug Addiction of Substance Abuse

According to World Health Organization (WHO), drugs are such substances which, when taken, change the normal behaviour of a living being.

A drug is not synonymous with narcotic substance. But we address narcotics as drugs in common day-to-day use of language. When a person develops a kind of physical and mental relationship with the drug due to continuous drug consumption and suffers from physical and mental problems if the drug is not taken regularly, it is called drug addiction or drug dependence. In other words, substance abuse. (Fig : 1.07).

The important drugs which create addiction are opium and opium products, heroin, wine, pathidin, barbiturate, cocaine, vung, choro, marijuana and LSD etc. Among them heroin is the most harmful. A man can be drug addicted for different causes e.g. curiosity, bad company, trying to remove depression, to get relief from mental stress, to make oneself more active, family feud and family habit and so on. If any of the parents is addicted to drugs, there is a possibility of drug addiction for the children.



Fig. 1.07 : Drug addiction destroys the life of a man completely.

1.6.1 Symptoms of Drug Addiction

There are some symptoms of an addicted person. These symptoms are not generally seen in a normal person. Important symptoms are-

- (i) Less attraction for food (ii) untidiness (iii) hazy vision and red eyes (iv) indifference to everything and disturbances in sleep (v) laziness and depression (vi) too much sweating (vii) staying away from others (viii) anxiety and tension (ix) less attentiveness, stealing money and other things from home.

Personal willingness or unwillingness certainly plays the most vital role in making an individual addicted to drugs. However, social and environmental influences are also instrumental in creating attraction for drugs among people.

The causes of drug addiction are mentioned in the table below:

Environmental causes	Family causes
i) Availability of drugs ii) Unemployment iii) Unsocial environment iv) Drop out from school early v) Drug business in the surrounding areas vi) Professional cause vii) Living in the place where antisocial activities or crimes take place frequently viii) Living in a place where opportunity to take drugs is rampant or where a group of addicts stay	i) Lack of control of parents ii) Depression iii) Loneliness iv) Indulgence in desperate behaviour of children v) Detachment from family vi) Carelessness to children (vii) Rough and violent livelihood or mentality viii) Bad company

1.6.2 Control of Drug Addiction

It is very difficult to stop drug addiction. Because a drug addict knows the destructive effects of drugs but cannot abstain from taking them. Drug addiction can be decreased by medical treatment, but in this case the role of the addicted person is very important. The person should be admitted to a drug cure centre or hospital and treatment should be done very systematically. At first the addicted person should be kept away from his addicted friends. He should be monitored so that drugs cannot come to his reach. Then he needs mental treatment so that he can forget the drug. For this he needs to get involved in a particular work. Drug addiction cannot be stopped suddenly. At first less amount of drug should be given to the addict for some days and then gradually giving drugs can be stopped. In this way, the bad effect of sudden stop of giving drugs can be decreased. Nerve relaxing medicines and sleeping pills can decrease the disturbance in sleep, anxiety and repulsive nature.

Drug addiction is not only a personal problem but also a great family problem and disturbance. This problem is a barrier to the development of society and country. Some dishonest people become rich by doing drug business, but on the other hand, life of some people is enveloped in darkness. Talented students are ruined and even die due to drug addiction. Social crimes also increase due to this. So, drug taking and its business should be strictly controlled. For controlling drugs the role of law enforcing agencies e.g. the role of government can be more effective than the effort from personal or social institutes.

Social Attempts

1. Finding out the addicted persons and providing medical treatment to them with compassion and without hatred.
2. Providing proper counselling to the addicts and their families by trained professionals.
3. Proper rehabilitation of the addicted persons and appropriate counseling of their families by trained personnel and getting them back to the normal stream of life in the society.

Government Attempts

1. Banning of taking and selling drugs by taking strict legal measures. This law should be enforced strictly.
2. The bad effects of drug addiction should be circulated by the government and non-government media.
3. It is worth mentioning that there is The Narcotics Control Act in our country. If the law is strictly applied, the people and the country can be saved from the harmful effects of drugs.

1.7 AIDS

AIDS is the most deadly disease in the world. It is a contagious disease. At first it was found in America in 1981. Since then it has been recognized as a deadly disease in the world. AIDS is mostly found in Africa.

Human beings have a natural system of immunity. There is a system in our blood which can protect us from diseases. In this case the lymphocytes of blood produce antibodies for the prevention of diseases. The persons who get this disease lose their immunity which causes death. So this disease is called AIDS (Acquired Immune Deficiency Syndrome). It is found that a virus called Human Immuno Deficiency Virus or HIV (Fig : 1.08) is the cause of this disease.

1.7.1 : Causes of AIDS

HIV destroys the natural immunity of the body. After entering into the blood stream of the body, HIV attacks the T-lymphocytes of White Blood Cells. For this reason, natural immune system of the body is destroyed. So the body is attacked with some rare diseases. Among them respiratory diseases, brain diseases, gastro intestinal diseases and tumours are worth mentioning. It has been observed that no disease symptoms are usually manifested in the human body for the first 5-10 years after HIV infection. These persons thus, act as a carrier of this disease, and then they can infect others.

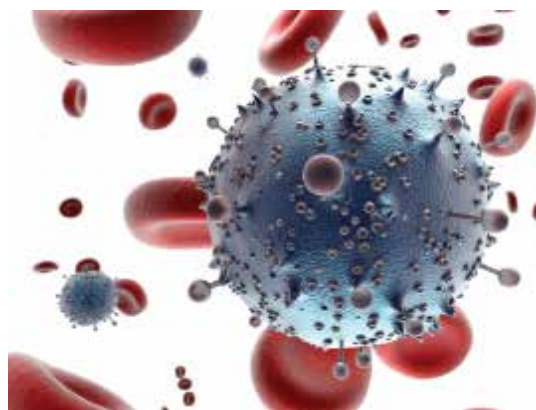


Fig. 1.08 : HIV virus of blood

A lot of information regarding AIDS has already been known to us. We now know who are most vulnerable to this disease. HIV is mainly transferred to healthy body by sexual intercourse. Unprotected sexual relationship either between men and women or homosexual relationships poses high risk for the transmission of this disease. If a pregnant woman is attacked with this disease, her child may also be attacked with the same. Newborn babies can get this disease by breast-feeding. Besides, the disease can be transmitted by blood transfusion from the diseased person. The use of same syringe by multiple drug addicts may also transmit the disease. This disease cannot be transmitted by food, water, insects or external touch of the body of the AIDS patients. This disease is most probably transmitted via blood, sperm etc body fluids.

The most important thing about the prevention of the disease is to inform people about how AIDS is transmitted. The spreading of this disease can be decreased by not infecting others and protecting oneself from contagion. The knowledge about the unprotected sexual relations, being conscious of the dangers of using the syringe of drug users, being careful about donating and receiving blood can reduce the risk of this disease. The government and social organizations can create public awareness for the prevention of this disease. Recently, bone marrow transplants have experimentally cured some AIDS patients completely. Apart from this, the disease can be kept under control by using oral medication, although it is not completely cured.

1.8 Physical Exercise and Rest for Health

Body is the first impression of man. So, body is called the most important tool for the struggle of life. It is our responsibility to keep this tool in proper condition. Regular physical exercise is essential for a healthy body. Physical fitness also develops from physical exercise.

Sleep, food and rest are essential for human life because these help the different parts of the human body to work properly. But it should be kept in mind that it does not develop the hidden treasure of the body. This development is possible only by taking regular exercise.

We know that the nervous system controls the muscles of our body. So, if we regularly exercise the muscles; it can make the nervous system fresh and active. Physical exercise activates the physiological and metabolic system of the body. So, our capacity for daily work increases. Taking exercise just for a few minutes can provide us with proper digestive capacity, blood circulation, good respiratory system and proper heat control of the body. That means we will get a healthy body. It should be remembered that the functions of the muscle have a great role in this matter. We have to take regular exercise so that the main muscles of the body become active and stimulated. The habit of physical exercise should be chosen depending on age, general health and body structure.

The habit of physical exercise should be developed considering age, physical structure, general health and so on. Exercise is not a matter of mere dismal labour. All kinds of sports are sources of recreation in one hand and they are types of physical exercise on the other. Nowadays, boys and girls equally take part in

games ranging from running, swimming, walking, cycling and karate to football, tennis, hokey and cricket. To keep our body fit, we should not only take exercise regularly but also lead our daily life accordingly. If a man walks 8000 to 10000 steps daily, he can hope to live a healthy, disease-free and longer life.

When a man becomes very tired of hard labour, then the muscles of the body become inactive, and so, we should keep our body in rest without work. This is called rest. Sleeping is the best rest. Daily six hours sleep is essential for keeping the body and mind fresh and healthy. Boys and girls need 8/9 hours and children need 10/12 hours sleep. The people who work at night need to get rest at day time.

Rest of Mind

Not only the body but also the mind needs rest. To remove all the stress and anxiety from the body and mind, both body and mind should have sound sleep. In this way the body and the mind gets proper rest.

It has been found that body and mind gets rest even by giving attention from one work to another. This is called rest by change of work. Recreation after hard labour is a rest, on the other hand, after hard mental labour doing a different work is the way to get rest.

It is seen that many famous writers are cleaning their fountain pen for hours together. He is getting rest doing this work. Some people take rest by gardening, rearing pets, doing amateur vegetable gardening or by recreation. All these are called rest by change of work.



Sample Questions



Multiple Choice Questions

1. What is mainly available in carrots?

- a. Phosphorus b. Vitamin D
- c. Carotene d. Vitamin B complex

2. Fat soluble vitamins in the body are

- i. A, D, E
- ii. A, B, C
- iii. A, D, K

Which of the following is correct?

- a. i and ii b. i and iii
- c. ii and iii d. i, ii and iii

Read the paragraph below and answer the questions 3 and 4.

Mahira likes to eat burgers, pizza, cake, soft drinks, and chips and she often consumes outside food. On the other hand, Anni prefers home-made meals, fresh fruits, vegetables, fish, and fiber-rich foods, and she takes them in proper quantities.

3. What describes Anni's food habit?

- a. rich in saturated animal fat
- b. tends to create frequent hunger
- c. maintains smooth and fresh skin
- d. causes mental stress and nervous weakness

4. Mahira's preferred foods—

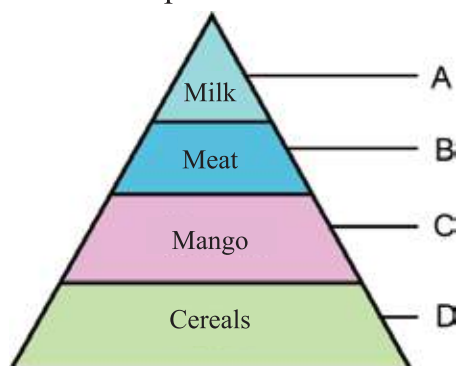
- i. may damage teeth and skin
- ii. may lead to vitamin and mineral deficiency in the body
- iii. may cause excess fat accumulation in the body

Which of the following is correct?

- a. i and ii b. i and iii
- c. ii and iii d. i, ii and iii

**Creative Question**

1. Tonu is 14 years old. Her weight is 35 kg and height is 1.5 meters. Recently, red patches have appeared on her skin, and she has little appetite. However, her body temperature is normal.
 - a. What are Nutrients?
 - b. What is meant by xerophthalmia?
 - c. Explain Tonu's health condition based on the BMI standard.
 - d. Analyze the solutions to Tonu's problems.
2. Look at the picture below and answer the questions:



- a. What is roughage?
- b. Which type of protein has higher nutritional value? Explain.
- c. Explain why food item A is placed at the top of the food pyramid.
- d. Although the food item marked D is nutritionally important, it needs to be taken required (small) quantity daily- analyze why.

**Short Answer Questions**

1. Why does the human body become obese?
2. What is the fat in fish called? Explain.
3. Why doesn't food spoil easily in winter?
4. Why is a "change of work" called a kind of rest? Explain.
5. Why is it better to eat less at night? Explain.